

Jake Amaral

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EDUCATION

University of Florida

Aug 2024 – May 2028

B.S. in Computer Science and Minor in Economics

Gainesville, FL

- **GPA:** 3.52
- **Relevant Coursework:** Data Structures & Algorithms, Prog in C++, Prog in Python, Discrete Structures
- **Extracurricular:** Kappa Sigma Fraternity, Association for Computing Machinery

TECHNICAL SKILLS

Languages: Python, C++, C#, SQL (PostgreSQL), HTML, TypeScript

Frameworks: Node.js, Next.js, React, Tailwind, shadcn, Drizzle ORM, AWS SAM

Developer Tools: Git/GitHub, Vercel, Supabase, Clerk, Unity, Google OAuth, AWS, Docker

PROFESSIONAL EXPERIENCE

Software Engineering Intern

May 2026 – Jul 2026

Safe Cities NPO | E-Portfolio

Cape Town, South Africa

- Led a 4-person team to build and ship a full-stack platform now used by **50+** nonprofit staff, using Next.js, TypeScript, SQL (PostgreSQL), and Drizzle ORM
- Enforced role-based access for every user by implementing Clerk authentication with Google OAuth 2.0 and Calendar API integrations
- Delivered a production-ready application from ambiguous requirements by interviewing stakeholders and iterating designs directly with Safe Cities staff
- Gained a broader perspective on global software development and cross-cultural collaboration through **direct cultural immersion**

Information Technology Support Assistant

Nov 2023 – Jun 2024

West Morris Mendham High School

Mendham, NJ

- Resolved **over 200** hardware and software tickets for students and staff, diagnosing issues, repairing devices, and maintaining a fleet-wide laptop inventory through an IT help desk system

PROJECTS

Spatial Math Interpreter | AWS (Bedrock, Lambda, IAM, SAM), Python, Docker

Sep 2026 – Present

- Designed and deployed a **serverless AWS backend** with AWS SAM, using Docker for containerized builds
- Paired a deterministic C# calculus engine with an LLM on Amazon Bedrock, ensuring the AI explains results but never calculates them
- Reduced operating costs and closed off attack surface through DynamoDB response caching, API-key usage plans with rate limits, least-privilege IAM roles, and input validation hardened against prompt injection

Spatial Math | C#, Unity

Apr 2026 – Present

- Built a real-time 3D graphing calculator that compiles user-typed equations into executable functions, supporting variables, nested functions, operator precedence, and implicit multiplication
- Rendered continuous, height-mapped surfaces by developing procedural mesh generation that samples equations across a configurable grid of **3,700+ vertices**, triangulated using Unity's core Mesh API
- Enabled real-time surface analysis by engineering a click-to-inspect feature: raycasting detects the selected point, and C# numerical differentiation computes and renders a matching 3D tangent plane

Enterprise Decision Making (EDM) AI Hackathon Entry | IBM watsonx Orchestrate

Feb 2026

- Designed a workflow of **6 agents** in IBM's watsonx AI Agent Builder to support business decision-making and governance at scale
- Surfaced inconsistencies and policy concerns hidden across **60+** synthetic data points by building specialized extraction agents that parsed a structured JSON dataset
- Preserved human authority over final outcomes by architecting an auditable review process, improving decision consistency and transparency without removing the human from the loop

LANGUAGES & INTERESTS

Languages: English (Native), Spanish (Intermediate High), Portuguese (Basic)

Hobbies: Baseball, Electronic Music Production, Golf, Martial Arts, Traveling